Example 1.5 (Simple N*-Body Molecular System)* We can now describe a molecular system consisting of N neutral atoms with pairwise Lennard-Jones potentials; this could be used to model liquid argon or a similar inert system; it is one of the first important simulations performed using a smooth classical potential [306, 387]. The Lennard-Jones potential is

$$\varphi_{\mathrm{LJ}}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right],$$

with typical parameters (for argon) $\epsilon = 1.65 \times 10^{-21}$ J and $\sigma = 3.4 \times 10^{-10}$ m. The energy of the system is

$$E = \frac{1}{2}\sum_{i=1}^{\mathrm{N}} m\dot{\boldsymbol{q}}_i^2 + \sum_{i=1}^{\mathrm{N}-1}\sum_{j=i+1}^{\mathrm{N}} \varphi_{\mathrm{LJ}}(r_{ij}),$$

where $r_{ij} = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|$ and the mass m of an argon atom is 6.69×10^{-26} kg. The equations of motion are, for $i = 1, 2, \ldots, \mathrm{N}$, using the chain rule,

$$\begin{aligned} m\ddot{\boldsymbol{q}}_i &= \sum_{\substack{j=1 \\ j\neq i}}^{\mathrm{N}} \frac{\varphi'_{\mathrm{LJ}}(r_{ij})}{r_{ij}}(\boldsymbol{q}_i - \boldsymbol{q}_j) \\ &= -24\frac{\epsilon}{\sigma}\sum_{\substack{j=1 \\ j\neq i}}^{\mathrm{N}} r_{ij}^{-1}\left[2\left(\frac{\sigma}{r_{ij}}\right)^{13} - \left(\frac{\sigma}{r_{ij}}\right)^{7}\right](\boldsymbol{q}_i - \boldsymbol{q}_j). \end{aligned}$$

Now introduce the change of variables

$$\boldsymbol{Q}_i = \sigma^{-1}\boldsymbol{q}_i, \qquad i = 1, 2, \ldots, \mathrm{N},$$

then $\dot{\boldsymbol{q}}_i = \sigma\dot{\boldsymbol{Q}}_i$, $\ddot{\boldsymbol{q}}_i = \sigma\ddot{\boldsymbol{Q}}_i$, and $\|\boldsymbol{q}_i - \boldsymbol{q}_j\| = \sigma\|\boldsymbol{Q}_i - \boldsymbol{Q}_j\|$. The equations of motion in these normalized coordinates may be written

$$\frac{m\sigma^2}{\epsilon}\frac{d^2\boldsymbol{Q}_i}{dt^2} = -\sum_{\substack{j=1 \\ j\neq i}}^{\mathrm{N}} \frac{\hat{\varphi}'_{\mathrm{LJ}}(R_{ij})}{R_{ij}}(\boldsymbol{Q}_i - \boldsymbol{Q}_j),$$

where $R_{ij} = \|\boldsymbol{Q}_i - \boldsymbol{Q}_j\|$ and $\hat{\varphi}_{\mathrm{LJ}}(R) = 4[R^{-12} - R^{-6}]$. Thus we see that if we make the additional time transformation

$$\tau = \alpha t, \qquad \alpha^2 = \frac{\epsilon}{m\sigma^2},$$

Quantity	Unit	Typical value (Ar)
length	σ	3.4×10^{-10} m
mass	m	6.69×10^{-26} Kg
energy	ϵ	1.65×10^{-21} J
time	$\sigma\sqrt{\mathrm{m}/\epsilon}$	2.17×10^{-12} s
velocity	$\sqrt{\epsilon/\mathrm{m}}$	1.57×10^{2} m/s
force	ϵ/σ	4.85×10^{-12} Newtons
temperature	$\epsilon/\mathrm{k_B}$	120K
pressure	ϵ/σ^3	4.20×10^{7} Newtons/m^2

Fig. 1.13 Table of units and their values for a model of argon (Ar)

then the equations become

$$\frac{d^2\boldsymbol{Q}_i}{d\tau^2} = -\sum_{\substack{j=1 \\ j\neq i}}^{\mathrm{N}} \frac{\hat{\varphi}'_{\mathrm{LJ}}(R_{ij})}{R_{ij}}(\boldsymbol{Q}_i - \boldsymbol{Q}_j).$$

This means that a natural choice for the unit of time is

$$\alpha^{-1} = \sigma\sqrt{\frac{m}{\epsilon}} \approx 2.17 \times 10^{-12}\mathrm{s}.$$

By using the units given here, we may work with a simplified form of the Lennard-Jones system involving unit masses and a parameter-independent potential energy function.

In Fig. 1.13 we summarize the typical units relevant for studies of argon. The role of temperature and pressure will be discussed in Chaps. 6–8. □

1.3 The Lagrangian and Generalized Coordinates

An N-body system is often formulated in terms of its *Lagrangian* $L = L(\boldsymbol{q}, \dot{\boldsymbol{q}})$: the difference between kinetic and potential energies. The somewhat mysterious Lagrangian function was introduced by Joseph Louis Lagrange. Its relevance becomes clear in formulating the *principle of least action*, a variational principle which gives an abstract formulation of the equations of motion. We defer introducing this derivation of the equations of motion until Chap. 2, where it is also used as a means of deriving a numerical method for molecular simulation.

For the system (1.3) with N atoms and $\mathrm{N_c} = 3\mathrm{N}$ configuration coordinates, the Lagrangian is

$$L \stackrel{\mathrm{def}}{=} \frac{\dot{\boldsymbol{q}}^T \boldsymbol{M} \dot{\boldsymbol{q}}}{2} - U(\boldsymbol{q}).$$